Exercise 2 - Approximation & Complexity theory

Problem 1 (Stone-Weierstrass theorem):
The following is a generalization of Weierstrass theorem presented in the lecture.

Theorem 1 (Stone-Weierstrass, see Theorem 2.2. in Telgarsky (2021)). *Let function class $\mathcal{F}$ of real-valued functions defined on $[0,1]^d$ be given as follows:*

i) Each $f \in \mathcal{F}$ is continuous.

ii) For every $\mathbf{x} \in [0,1]^d$, there exists $f \in \mathcal{F}$ with $f(\mathbf{x}) \neq 0$.

iii) For every $\mathbf{x} \neq \mathbf{x}' \in [0,1]^d$, there exists $f \in \mathcal{F}$ with $f(\mathbf{x}) \neq f(\mathbf{x}')$ ($\mathcal{F}$ separates points).

iv) $\mathcal{F}$ is closed under multiplication and vector space operations, i.e. $\mathcal{F}$ is an algebra.

Then $\mathcal{F}$ is a universal approximator: for every continuous $g : [0,1]^d \to \mathbb{R}$ and $\epsilon > 0$, there exists $f \in \mathcal{F}$ with

$$|f(\mathbf{x}) - g(\mathbf{x})| \leq \epsilon, \ \forall \mathbf{x} \in [0,1]^d.$$

a) Consider unbounded width networks with one hidden layer:

$$\mathcal{F}_{\sigma,d,m} := \mathcal{F}_{d,m} := \{x \mapsto a^\top \sigma(Wx + b) : a \in \mathbb{R}^m, W \in \mathbb{R}^{m \times d}, b \in \mathbb{R}^m\}. \quad (1)$$

$$\mathcal{F}_{\sigma,d} := \mathcal{F}_d := \bigcup_{m \geq 0} \mathcal{F}_{\sigma,d,m}. \quad (2)$$

a) To show that $\mathcal{F}_{\cos,d}$ is universal, we can show that it fulfils all properties of the Stone-Weierstrass Theorem:

(i) each $f \in \mathcal{F}_{\cos,d}$ is continuous:

yes, since $\cos(\cdot)$ is a continuous function & composition of continuous functions via addition & (vector)-multiplication remains continuous.

(ii) $\forall x \in [0,1]^d \ \exists f \in \mathcal{F}_{\cos,d} : f(x) \neq 0$

yes, since we can construct such a function in the following way:

choose $\underline{a} = \underline{1}$ the vector of all one's.

$\underline{\underline{W}} = \underline{\underline{0}}$ to be a matrix of all zeros

and $\underline{b} = \underline{0}$ the zero vector.

Then for any $x \in [0,1]^d$: $f(x) = \underline{1}^\top \cos(\underline{\underline{0}} \cdot x + \underline{0}) = \underline{1}^\top \underline{1} = d \neq 0$

(iii) $\forall x, x' \in [0,1]^d,\ x \neq x'\ \exists f:\ f(x) \neq f(x')$

w.l.o.g. consider $m = 1$, that is $a \in \mathbb{R}$, $b \in \mathbb{R}$, $W \in \mathbb{R}^{1 \times d}$.

While $\cos(x)$ is a periodic function, we note that for any $x, y \in [0, 2\pi)$ with $x \neq y$, we have $\cos(x) \neq \cos(y)$.

To simplify the construction of f, let us choose $a = 1$ and $b = 0$.

Then we just need to show that

for any $x, x' \in [0,1]^d$ w/ $x \neq x'$ we can find a $W \in \mathbb{R}^{1 \times d}$, such that

$$\cos(Wx) \neq \cos(Wx').$$

One possible way to construct it: if $x \neq x'$, it means that they differ in at least one entry $i \in \{0, 1, \dots, d\}$.

Choose $W = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = e_i$, then $Wx = x_i \neq x_i' = Wx'$ and $x_i, x_i' \in [0,1]$.

Then $\cos(Wx) = \cos(x_i) \neq \cos(x_i') = \cos(Wx')$ and

$f(x) = 1 \cdot \cos(e_i \cdot x + 0) \neq 1 \cdot \cos(e_i \cdot x' + 0) = f(x_i)$.

Note: For different x, x' the function f will also differ! What matters is that we can find such a function $f \in \mathcal{F}_{\cos, d}$ for any $x \neq x'$.

(iv) $\mathcal{F}$ is an algebra, i.e. $\forall f_1, f_2 \in \mathcal{F}$

① $\alpha f_1 + \beta f_2 \in \mathcal{F}$

② $f_1 \cdot f_2 \in \mathcal{F}$

1) Let $f(x) = a^T \sigma(Wx + b)$; $\alpha, \beta \in \mathbb{R}$

$g(x) = a'^T \sigma(W'x + b')$

$h(x) =$

$$\alpha f(x) + \beta g(x) = [\alpha a^T \ \beta a'^T] \, \sigma\left(\begin{bmatrix} W \\ W' \end{bmatrix} x + \begin{bmatrix} b \\ b' \end{bmatrix}\right) = \tilde{a}^T \sigma(\tilde{W}x + \tilde{b}) \in \mathcal{F}_{\cos, d} \ \checkmark$$

2) We will use the trigonometric rule: $\cos(\alpha)\cdot\cos(\beta) = \frac{1}{2}(\cos(\alpha-\beta)+\cos(\alpha+\beta))$

Then

$$f(x)\cdot g(x) = \left(\sum_{i=1}^{m} a_i \cos(w_i x + b_i)\right)\cdot\left(\sum_{j=1}^{m'} a_j' \cos(w_j' x + b_j')\right)$$

$$= \sum_{i=1}^{m}\sum_{j=1}^{m'} \frac{a_i a_j'}{2}\left(\cos\left((w_i x+b_i)-(w_j' x+b_j')\right) + \cos\left((w_i x+b_i)+(w_j' x+b_j')\right)\right)$$

$$= \underbrace{\sum_{i=1}^{m}\sum_{j=1}^{m'} \frac{a_i a_j'}{2}\cdot\cos\left((w_i - w_j')x + (b_i - b_j')\right)}_{h_1(x)\in \mathcal{F}_{\cos,1}} + \underbrace{\sum_{i=1}^{m}\sum_{j=1}^{m'} \frac{a_i a_j'}{2}\cdot\cos\left((w_i + w_j')x + (b_i + b_j')\right)}_{h_2(x)\in \mathcal{F}_{\cos,1}}$$

$\Rightarrow$ using 4): $h_1(x)+h_2(x)\in \mathcal{F}_{\cos,1}$ ✓

b) Now consider the following definitions of function classes. Check whether each of the necessary conditions in Theorem 1 hold and argue whether $\mathcal{F}$ is dense in $C[0,1]$

i) Let $\mathcal{F} = \text{span}\{x\} = \{f : [0,1] \to \mathbb{R} | f \text{ linear}\}$ be the function class of all linear functions.

ii) Let $\mathcal{F} = \{f \text{ is polynomial} \mid f(0) = 0\}$ be the function class $\mathcal{F}$ of all polynomials with no constant term.

iii) Consider now the interval $[0, 2\pi]$. (We can always define a projection function $\phi(x) : [0, 2\pi] \to [0, 1]$, to map the interval to the unit interval.)
Let $\mathcal{F} = \{a_0 + \sum_{k=1}^{n}(a_k \cos(kx) + b_k \sin(kx)) | n \in \mathbb{N}, a_k, b_k \in \mathbb{R}\}$, the function class containing all real trigonometric polynomials.

iv) For any subset $S \subset [0,1]$, define the characteristic function $\mathbf{1}_S : x \mapsto \begin{cases} 1 & , x \in S \\ 0 & , x \notin S \end{cases}$.
Let
$$\mathcal{F} := \text{algebra}\{\mathbf{1}_{\{a\}} | a \in [0,1]\} = \{f | \text{supp}(f) \text{ is finite}\}.$$
where the support $\text{supp}(f)$ of a function f is defined as $\text{supp}(f) := \{x \in [0,1] | f(x) \neq 0\}$.

b) To answer whether $\mathcal{F}$ is dense in $C([0,1])$, we don't need to check all conditions. It is sufficient if at least on Property does not hold. Checking all conditions is just an exercise for you.

(i) All linear functions $f \in \mathcal{F}$ can be expressed as $f(x) = mx + b$ for some $m, b \in \mathbb{R}$.

(i) holds because linear funct. are continuous func. ✓

(ii) for any $x \in [0,1]$ we can just choose $f(x) = mx + b = b \neq 0$ ✓
by choosing $m = 0$, $b \neq 0$.

(iii) for any $x \neq x'$, we can choose $f(x) = mx + b$ with $m \neq 0$, $b = 0$ ✓
then $f(x) = mx \neq mx' = f(x')$.

(iv) $\mathcal{F}$ is not an Algebra because it is not closed under multiplication. ↯
for $f, g \in \mathcal{F}$, $f(x) = mx + b$
$g(x) = m'x + b'$

$$f\cdot g = (mx+b)\cdot(m'x+b') = mm'x^2 + (mb'+m'b)x + bb'$$

(quadratic term ↯; $+bb'$ well.)

$\mathcal{F}$ is not dense in $C([0,1])$. (It cannot approx. polynomials arbitrarily well.)

(ii) F consists of all functions of the form: $f(x) = \sum_{i=1}^{n} a_i \cdot x^i$ for $n \in \mathbb{N}$, $a_i \in \mathbb{R}$ (no constant term!)

(i) polynomials are continuous functions ✓

(ii) for $x = 0$ $f(x) = 0$ for all $f \in F$ by construction. ↯ → F is not dense in $C[0,1]$

(iii) for $x \neq x'$, we can just the same example as above: e.g. $f(x) = x$

(iv) for $f, g \in F$ we have (w.l.o.g assume $n \geq m$) then $f(x) = x \neq x' = f(x')$ by assumption.

$$f(x) = \sum_{i=1}^{n} a_i x^i \,, \quad g(x) = \sum_{j=1}^{m} b_j x^j \,,$$

$$f(x) + g(x) = \sum_{i=1}^{m} (a_i + b_i) x_i + \sum_{j=m+1}^{n} a_j x^j \quad ✓$$

$$(f \cdot g)(x) = \left(\sum_{i=1}^{n} a_i x^i\right) \cdot \left(\sum_{j=1}^{m} b_j x^j\right) = \ldots$$

$n \cdot m$ summands, for any summand we have: $a_i x^i \cdot b_j x^j = (a_i b_j) \cdot x^{i+j}$ with $i + j \geq 1$

→ can be rewritten as $h(x) = \sum_{k=1}^{n+m} c_k x^k$ ✓

→ F is not dense because it cannot approximate constant functions $c(x) = c, c \neq 0$ at $x = 0$.

(iii) $f \in F$: $f = a_0 + \sum_{k=1}^{n} a_k \sin(kx) + b_k \cos(kx)$, $n \in \mathbb{N}$, $a_k, b_k \in \mathbb{R}$

(i) cos & sin are continuous func. ✓

(ii) choose $n = 1$: $f(x) = a_0 + a_1 \sin(x) + b_1 \cos(x)$

choose $a_1 = b_1 = 1$, then $|a_1 \sin(x) + b_1 \cos(x)| \leq 2$

→ choosing $a_0 > 2$ guarantees that $f(x) \neq 0$ $\forall x \in [0, 2\pi)$

(iii) for $x \neq x'$: note that $\sin(x) = \cos(x + \frac{\pi}{2})$ and the symmetry along for $(|x - \frac{\pi}{2}| = |x' - \frac{\pi}{2}|) \vee (|x - \frac{3\pi}{2}| = |x' - \frac{3\pi}{2}|)$: $\cos(x) = -\cos(x')$ else : $\sin(x) \neq \sin(x')$ $\left\{ \begin{array}{l} \frac{\pi}{2} \text{ for the interval } [0, \pi) \\ \frac{3}{2}\pi \text{ for the interval } [\pi, 2\pi) \end{array} \right.$ for $\sin(x)$.

(iv) F is an algebra following similar reasons as in a):

Note that $\sin(kx) \cdot \cos(\ell \cdot x) = \frac{1}{2}\left(\sin(kx + \ell x) + \sin(kx - \ell x)\right)$

$\sin(kx) \cdot \sin(\ell \cdot x) = \frac{1}{2}\left(\cos(kx - \ell x) - \cos(kx + \ell x)\right)$

$$\cos(kx)\cdot\cos(\ell x) = \frac{1}{2}\left(\cos(kx+\ell x) + \cos(kx-\ell x)\right)$$

$\Rightarrow \mathcal{F}$ is dense in $C([0,1])$

(iv) $\mathcal{F} := \text{algebra}\{\mathbf{1}_{\{a\}} \mid a\in[0,1]\} = \{f \mid \text{supp}(f) \text{ is finite}\}$



$\text{supp}(f) := \{x\in[0,1] \mid f(x)\neq 0\}$.

with subset $S\subset[0,1]$ and the characteristic function $1_S : x \mapsto \begin{cases} 1, & x\in S \\ 0, & x\notin S \end{cases}$

Then (i) : · some (almost all) functions in $\mathcal{F}$ are not continuous. ↯

(ii) · for each $x\in[0,1]$, $1_{\{x\}}\in\mathcal{F}$ satisfies $1_{\{x\}}(x) = 1 \neq 0$

(iii) · for each $x\neq x'\in[0,1]$, $1_{\{x\}}(x) = 1 \neq 0 = 1_{\{x\}}(x')$

(iv) · by definition, $\mathcal{F}$ is an algebra.

But $\mathcal{F}$ is clearly not dense in $C([0,1])$.

For instance: $|1_{[0,1]}(x) - f(x)| = 1 \quad \forall x\in[0,1]\setminus\text{supp}(f)$.

$\Rightarrow \mathcal{F}$ is not dense in $C([0,1])$

p. 73, Lemma 22

Problem 2 (Number of affine pieces in a ReLU network):
We will prove the lemma that bounds the number of affine pieces in a ReLU network stated in the lecture notes. Denote by $\delta_A(f)$ be the (minimum) number of affine pieces of a piece-wisely linear continuous function $f : \mathbb{R} \to \mathbb{R}$. Show that

a) $\delta_A(\sum_k a_k f_k + b_k) \leq \sum_k \delta_A(f_k)$, for any finite sequence of piece-wisely linear continuous functions f_k, and for any real sequences a_k and b_k.

b) Let $g_i : \mathbb{R} \to \mathbb{R}$ denote the output of some node in layer i in a ReLU network with L layers of widths $(m_1, \ldots, m_L)$ as a function of the input. Using induction on i, show that the number of affine pieces $\delta_A(g_i)$ satisfies

$$\delta_A(g_i) \leq 2^i \prod_{j<i} m_j.$$

c) Recall that $g_L : \mathbb{R} \to \mathbb{R}$ is a ReLU network with L layers of widths $(m_1, \ldots, m_L)$ such that $m = \sum_{i=1}^{L} m_i$. The number of affine pieces in g_L satisfies

$$\delta_A(g_L) \leq \left(\frac{2m}{L}\right)^L.$$

d) What value of L maximizes the upper bound $\left(\frac{2m}{L}\right)^L$? Although taking this value of L gives a large value of $\delta_A(g_L)$, give a reason why in practice we usually pick a value much smaller than this.

a) from Prop. 32 from lecture notes we know:

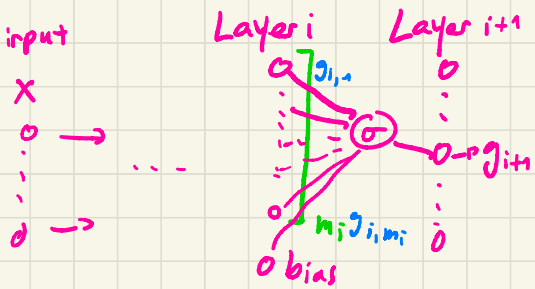
(i) $\delta_A\left(\sum_k f_k\right) \leq \sum_k \delta_A(f_k)$

Furthermore, note that $\delta_A(a_k f_k) \leq \delta_A(f_k)$ (strict ineq. for $a_k = 0$).

Finally note that δ_A does not change w/ constants, from which we conclude:

$$\delta_A\left(\sum_k a_k f_k + b_k\right) = \delta_A\left(\sum_k a_k f_k\right) \leq \delta_A\left(\sum_k f_k\right) \leq \sum_k \delta_A(f_k) \quad \square$$

b)



Proof by induction:

induction start: identity mapping at layer 0

$$\delta_A(g_0) = 1 \leq 2^0 \prod_{j<0} m_j = 1 \cdot 1 = 1 \quad \checkmark$$

induction step:

Assume $\delta_A(g_i) \leq 2^i \prod_{j<i} m_j$ holds for layer i.

Then: $\delta_A(g_{i+1}) = \delta_A\left(\sigma\left(b_i + \sum_{j=1}^{m_i} a_j g_{i,j}\right)\right) \leq 2\,\delta_A\left(b_i + \sum_{j=1}^{m_i} a_j g_{i,j}\right)$

$$\overset{(a)}{\leq} 2 \sum_{j=1}^{m_i} \delta_A(g_{i,j})$$

use induct. assumption

$$= 2 \sum_{j=1}^{m_i} 2^i \cdot \prod_{k<i} m_k$$

does not dep. on j

$$= 2 \cdot m_i \cdot 2^i \cdot \prod_{k<i} m_k$$

$$\delta_A(g_{i+1}) = 2^{i+1} \prod_{k<i+1} m_k \qquad \square$$

most affine pieces if hidden neurons are spread evenly over all layers

b) show: $\delta_A(g_L) \leq 2^L \prod_{j \leq L} m_j \leq \left(\frac{2m}{L}\right)^L$

show: $\prod_{j \leq L} m_j \leq \left(\frac{m}{L}\right)^L$

$$\prod_{j \leq L} m_j = \exp\left(\sum_{j \leq L} \ln(m_j)\right)$$

$$= \exp\left(L \sum_{j \leq L} \tfrac{1}{L} \ln(m_j)\right)$$

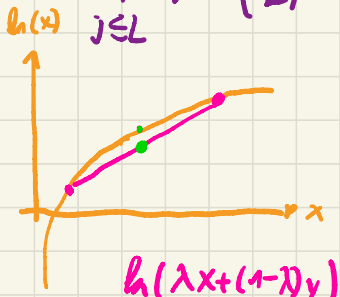
concavity of $\ln(\cdot)$ ↘

$$\leq \exp\left(L \cdot \ln\left(\sum_{j \leq L} \tfrac{1}{L} m_j\right)\right)$$

$$= \exp\left(L \cdot \ln\left(\tfrac{1}{L} \underbrace{\sum_{j \leq L} m_j}_{=m}\right)\right)$$

$$= \exp\left(L \cdot \ln\left(\tfrac{m}{L}\right)\right)$$

$$= e^{\ln\left(\frac{m}{L}\right) \cdot L} = \left(e^{\ln\left(\frac{m}{L}\right)}\right)^L = \left(\frac{m}{L}\right)^L \quad \square$$



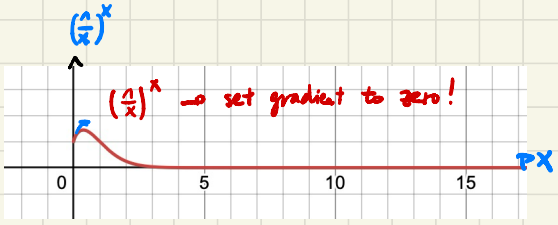
$\ln(\lambda x + (1-\lambda) y) \quad \lambda \in [0,1]$

$\geq \lambda \ln(x) + (1-\lambda) \ln(y)$

d) → look at function $\left(\frac{1}{x}\right)^x$ and find its maximum → why?

$\left(\frac{2m}{L}\right)^L = \left(\left(\frac{2m}{L}\right)^{\frac{L}{2m}}\right)^{2m}$

$\left(\frac{1}{x}\right)^x$ w/ $x = \frac{L}{2m}$



$$\frac{d}{dx}\left(\frac{1}{x}\right)^x = \frac{d}{dx}\left(e^{\ln\left(\frac{1}{x}\right)\cdot x}\right) = \ldots = \left(\frac{1}{x}\right)^x \cdot \underbrace{\left(\ln\left(\frac{1}{x}\right) - 1\right)}_{\ln\left(\frac{1}{x}\right)-1 \stackrel{!}{=} 0} \stackrel{!}{=} 0$$

$$\ln\left(\frac{1}{x}\right) - 1 \stackrel{!}{=} 0$$

$$1 = -\ln(x)$$

$$x = e^{-1}$$

$$\frac{L}{2m} = e^{-1} \Leftrightarrow L = 2e^{-1}m \sim m$$

$\Downarrow$

L grows prop. to m,

$\Rightarrow$ very deep networks

! vanishing gradients !